Short Literature Review: Predictive Methods for Weather and PM10 Data

Purpose

This report summarises three predictive methods suitable for the internship weather-station and PM10 datasets. The methods are ordered from easiest to hardest to implement:

1. Multiple Linear Regression / Regularised Regression
2. Random Forest or Gradient-Boosted Trees
3. LSTM / Deep Sequence Models

Dataset Context

The available project material contains weather measurements and PM10 measurements across several tasks:

- Task 1: 2026 weather station data and exploratory data analysis.
- Task 2: 2023-2026 meteorological station data, including yearly EDA reports and all-years summaries.
- Task 3 Part 1: monthly, seasonal, and yearly weather profiles.
- Task 3 Part 2: PM10 data aligned with Task 1 weather data.

The existing analysis material suggests several modelling-relevant patterns:

- PM10 has useful relationships with wind gust, wind speed, relative humidity, temperature, barometric pressure, solar radiation, and precipitation.
- In the PM10 analysis, wind gust and wind speed had the strongest same-hour relationship with PM10, both negative.
- Top 10% PM10 hours were associated with lower wind speed than other hours.
- Weather data is time-indexed, with repeated 10-minute or hourly patterns, so lagged values and time-of-day/seasonal features will likely matter.

Method 1: Multiple Linear Regression / Regularised Regression

Difficulty: Easy

Best use:

- First baseline model.
- Predicting next-hour or next-day PM10 from current weather variables.
- Predicting continuous weather variables such as temperature, humidity, wind speed, or solar radiation.

This method models the target as a weighted sum of predictors. For example, PM10 could be predicted using wind speed, wind gust, humidity, temperature, pressure, rainfall, solar radiation, hour of day, month, and lagged PM10 values. Ridge or Lasso regression can be used if several predictors are correlated.

Why it suits this project:

- Easy to explain in a short report.

- Fast to implement in Python.
- Produces interpretable coefficients, showing whether each variable is positively or negatively related to PM10.
- Matches published PM10 work using meteorological predictors. A Malaysian PM10 study built multiple linear regression models using meteorological factors and reported seasonal R2 values around 0.57-0.68, with wind speed, relative humidity, and rainfall negatively related to PM10, while temperature and pressure were positively related.

Limitations:

- Assumes mostly linear relationships.
- Struggles with sharp pollution spikes unless lagged PM10 and event indicators are included.
- May underperform when interactions matter, for example low wind speed plus high humidity plus certain wind direction sectors.

Recommended implementation:

- Target: next-hour PM10 or daily mean PM10.
- Features: current weather variables, previous PM10, previous 1-3 hour rolling means, hour, day of week, month/season, rain/no-rain flag, wind sector.
- Evaluation: chronological train/test split, MAE, RMSE, R2, and comparison against a persistence baseline.

Method 2: Random Forest or Gradient-Boosted Trees

Difficulty: Medium

Best use:

- Practical main model for tabular weather and PM10 prediction.
- Capturing nonlinear relationships and interactions.
- Ranking feature importance.

Tree ensemble models split the data into many decision rules and combine multiple trees to improve prediction. Random Forest is usually easier and more robust. Gradient boosting methods such as XGBoost, LightGBM, or scikit-learn HistGradientBoosting can be more accurate but need more tuning.

Why it suits this project:

- Handles nonlinear PM10-weather relationships better than linear regression.
- Works well with mixed variables: numeric readings, hour/month indicators, wind-sector categories, lag features, rolling averages, and rain flags.
- Gives feature importance, which helps explain which weather variables influence PM10.
- Literature supports this direction. A PM10 tree-based study found Random Forest was the best model for next-day PM10 concentration classification, with wind speed and temperature among the most influential variables. A weather-forecasting survey also identifies Random Forest and other ensemble methods as widely used traditional machine-learning methods for meteorological prediction.

Limitations:

- Less transparent than linear regression.
- Does not automatically understand temporal order; lag and rolling-window features must be engineered.

- Can overfit if the model is too deep or if train/test splitting is random instead of time-based.

Recommended implementation:

- Start with Random Forest Regressor for PM10 concentration.
- Also test Random Forest Classifier for high-PM10 events, such as top 10% PM10 hours.
- Use lag features: previous PM10, previous wind speed/gust, 3-hour rolling mean, 24-hour rolling mean, and recent precipitation.
- Use chronological validation, not shuffled random validation.
- Metrics:
  - Regression: MAE, RMSE, R2.
  - High-PM10 classification: precision, recall, F1, confusion matrix.

Method 3: LSTM / Deep Sequence Models

Difficulty: Hard

Best use:

- Advanced sequence forecasting.
- Predicting PM10 or weather variables from previous time windows.
- Modelling temporal dependence where the recent sequence matters more than individual rows.

Long Short-Term Memory models are recurrent neural networks designed for sequence data. Instead of predicting from one row at a time, an LSTM can use a window such as the previous 24 hours or previous 7 days of weather and PM10 readings to predict the next hour/day.

Why it suits this project:

- The data is naturally time-series data.
- PM10 and weather variables often depend on recent history, not just current conditions.
- Prior research has used LSTM for PM10 forecasting by converting PM10 data into sequence windows. One PM10 LSTM study used moving-average smoothing, sliding windows, and RMSE evaluation, and reported better performance than linear regression and a standard recurrent neural network.
- Recent air-quality forecasting literature also reports strong use of LSTM, Bi-LSTM, and hybrid deep-learning models with meteorological variables for PM2.5 and PM10 forecasting.

Limitations:

- Harder to implement and explain.
- Needs careful preprocessing, scaling, sequence-window construction, and train/validation/test splitting.
- More prone to overfitting if the dataset is small, especially for PM10 where the aligned Task 3 hourly table appears much smaller than the full weather dataset.
- Feature importance is less straightforward, so additional explainability methods may be needed.

Recommended implementation:

- Use only after building linear regression and Random Forest baselines.
- Build supervised sequences from hourly data:
  - Input window: previous 24 or 48 hours of PM10 + weather variables.

    - Prediction target: next-hour PM10, next-day mean PM10, or high-PM10 probability.
- Scale numeric variables using training data only.
- Compare against persistence, linear regression, and Random Forest.
- Metrics: MAE, RMSE, R2, and event recall for high-PM10 periods.

Recommended Order for This Project

The best workflow is not to choose the hardest model first. A good modelling plan would be:

1. Build a persistence baseline.
    - Example: predict next-hour PM10 equals current PM10.
2. Build the easy method: linear or regularised regression.
    - This gives an explainable first model and a useful comparison point.
3. Build the medium method: Random Forest / Gradient-Boosted Trees.
    - This is likely the best balance of accuracy, effort, and explainability for the internship dataset.
4. Consider the hard method: LSTM.
    - Use it only if the task requires sequence modelling or if simpler models cannot capture the patterns.

Final Recommendation

For the next practical modelling task, Random Forest or Gradient-Boosted Trees should be the main candidate. It is more powerful than linear regression, easier to implement and explain than LSTM, and well suited to the current weather + PM10 feature set.

Linear regression should still be included as the baseline because it is transparent and quick. LSTM should be described as the advanced option, and only implemented if there is enough time to prepare sequence windows and validate it properly.

References

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